

KernelAscent: RSI benchmark — figure atlas

September 15, 2026

Auto-generated figure atlas (regenerate: `python3 scripts/make_paper_figures.py`). Pick the figures that matter; each has a `label` for cross-referencing.

1 Leaderboards

Model	tier	rounds	$L - F$
starcoder2-15b	large	6	0.232
deepseek-coder-6.7b-base	mid	3	0.154
starcoder2-15b	large	5	0.071
deepseek-coder-1.3b-base	small	12	0.039
deepseek-coder-1.3b-base	small	8	0.038
deepseek-coder-1.3b-base	small	4	0.038
deepseek-coder-1.3b-base	small	24	0.001
Qwen2.5-Coder-3B	small	2	0.000
Qwen2.5-Coder-14B	large	4	-0.000
Qwen2.5-Coder-1.5B	small	3	0.000
Qwen2.5-Coder-1.5B	small	3	0.000
Qwen2.5-Coder-1.5B	small	12	0.000

Table 1: Task 5 open self-play: author co-evolution $L - F$ leaderboard.

Model	Q_0	Q_g	Δ vs base
us.openai.gpt-6-astra	0.298	0.854	0.556
us.anthropic.claude-sonnet-5	0.493	0.894	0.401
mistral.mistral-large-3-675b	0.359	0.669	0.310
Nova-Pro	0.401	0.490	0.089
us.openai.gpt-5.6-sol	0.813	0.889	0.075
Kimi-K2.5	0.752	0.813	0.062
Claude-Opus-5	0.878	0.910	0.032
Fable-5.1	0.732	0.741	0.009
us.openai.gpt-5.6-terra	0.851	0.850	-0.001
Llama-3.3-70B	0.458	0.455	-0.004
deepseek.v3.2	0.847	0.614	-0.234

Table 2: Task 3 procedure-RSI: held-out Q gain vs. frozen procedure.

Size band	n	P(cross wall)	P(RSI)	mean drift
<2B	26	0.54	0.04	0.024
2-8B	17	0.94	0.41	0.299
$\geq 9B$	7	0.71	0.29	0.396

Table 3: Scale $\rightarrow$ RSI gates: correctness-wall crossing and compounding by size band.

2 Mechanistic interpretability

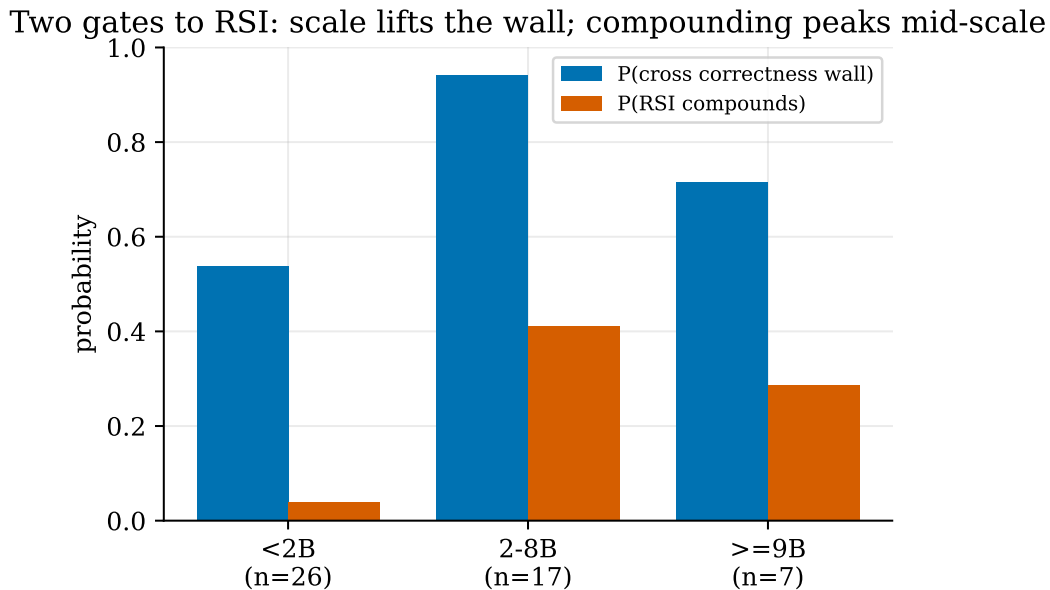


Figure 1: Scale $\rightarrow$ RSI. Below $\sim 2B$ the correctness wall is rarely crossed, so weight-RSI is impossible; crossing rises sharply at $2B+$, but compounding peaks at mid-scale ($2-8B$) and does not increase further $\geq 9B$ (headroom ceiling).

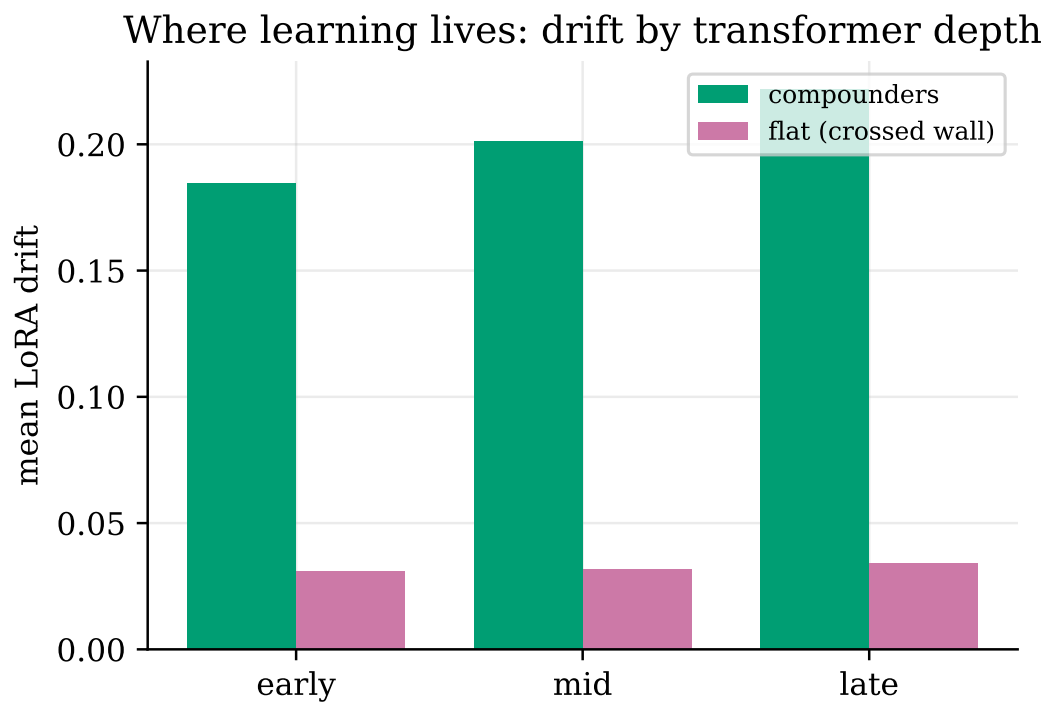


Figure 2: LoRA weight drift by transformer depth for compounds vs. flat (wall-crossing) runs. Compounds sustain markedly larger drift across all depths; flat runs saturate early.

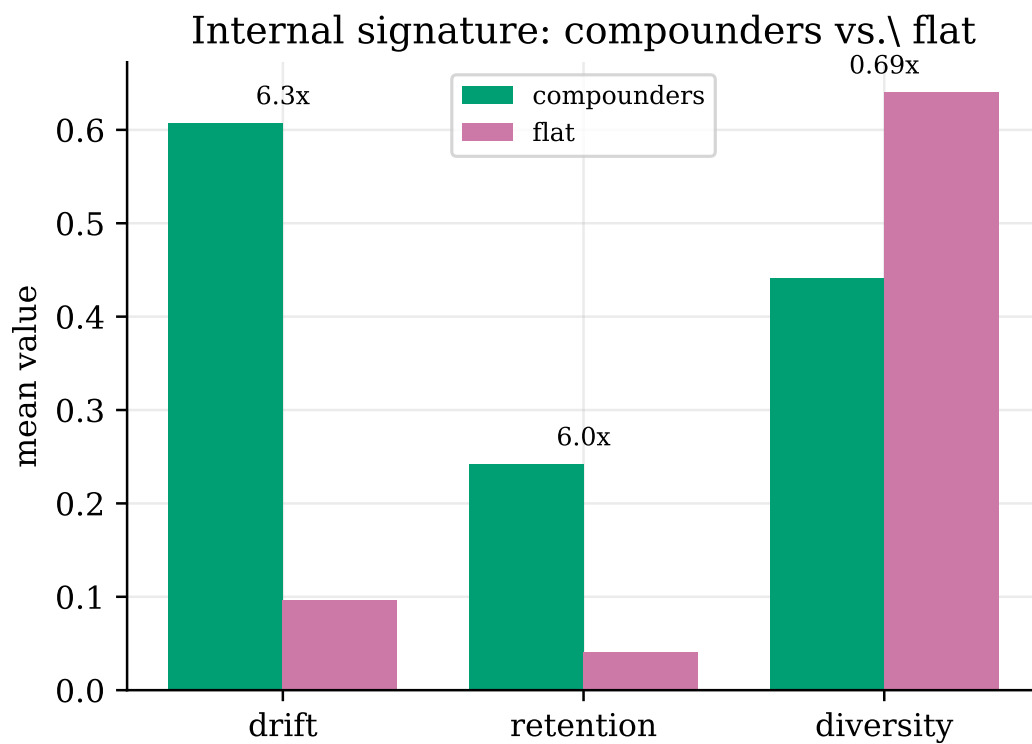


Figure 3: The internal signature that separates compounders from flat runs among wall-crossers: sustained drift and retention are $\sim 5\text{--}6\times$ higher in compounders, while generation diversity is *lower* (productive convergence, not diversity preservation).

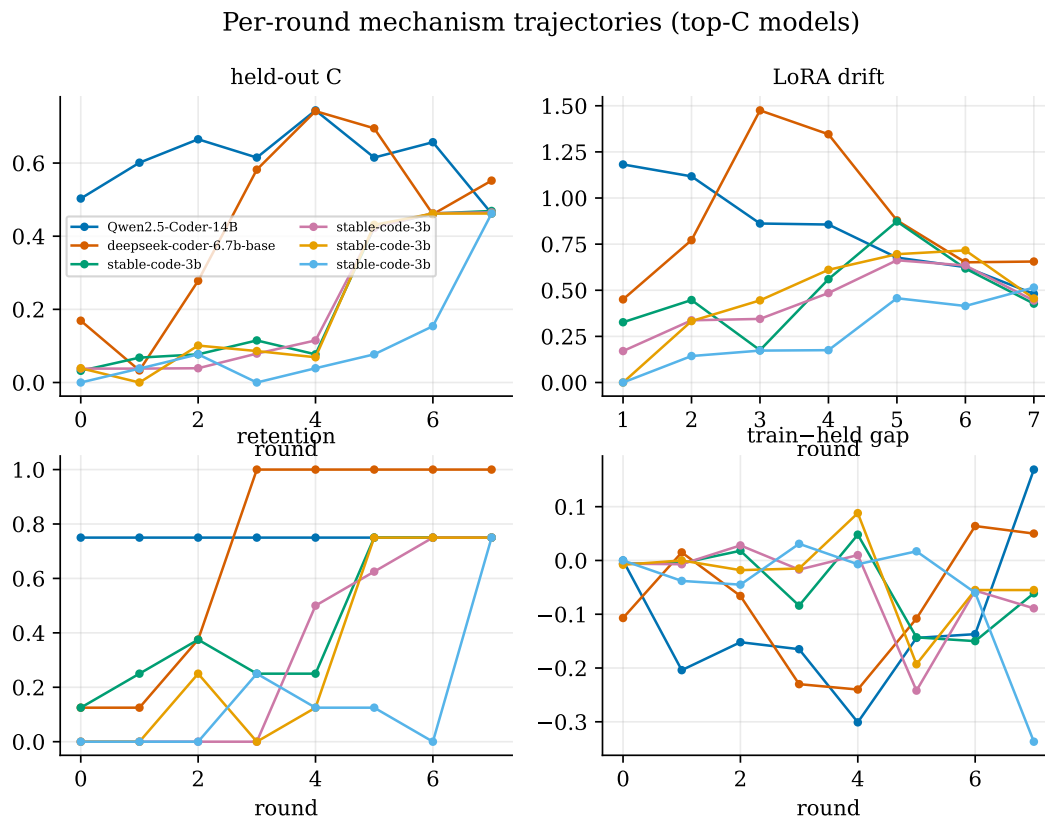


Figure 4: Per-round mechanism trajectories for the highest-capability probes: held-out C, total LoRA drift, retention, and the train–held transfer gap. Compounders show rising/held C with sustained drift and retention; flat runs plateau with drift saturation.

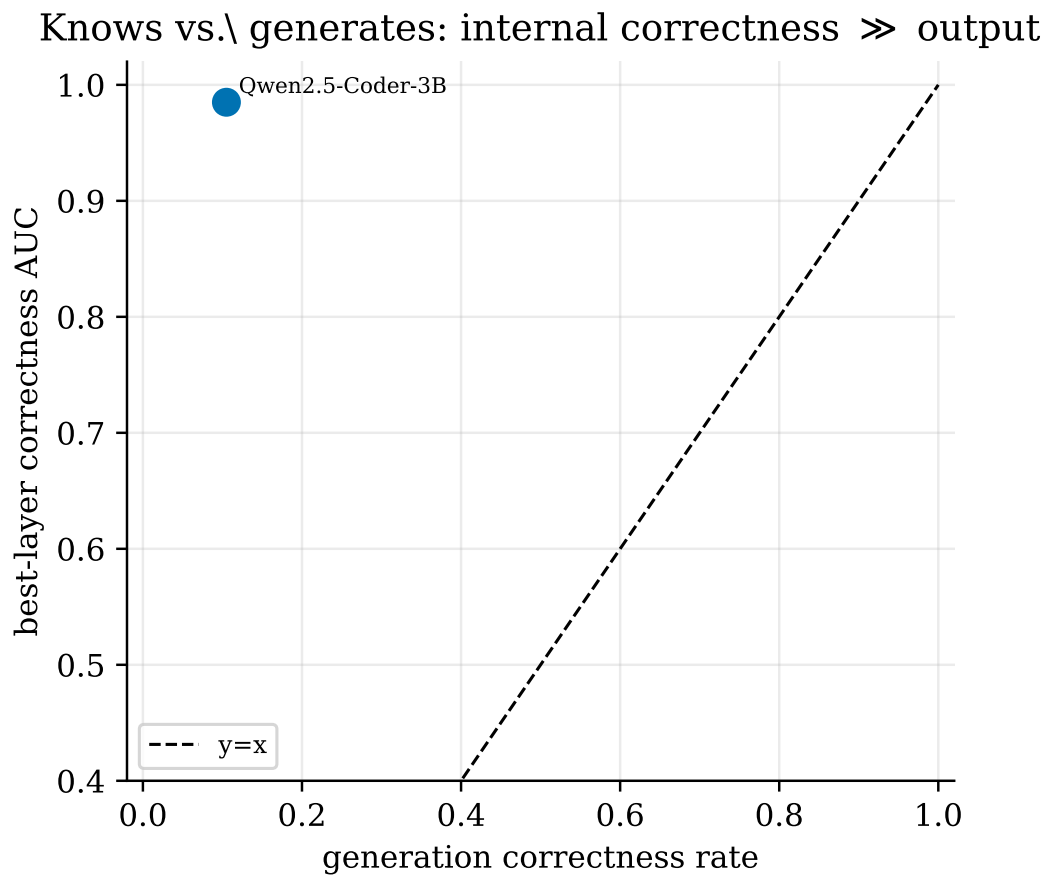


Figure 5: Internal correctness decodability (best-layer linear-probe AUC) vs. how often the model actually generates a correct kernel. Points far above $y = x$ mean the model *represents* correctness internally far better than it can *decode* it into a correct kernel: the RSI bottleneck is generation, not knowledge.

3 Task 5: self-play

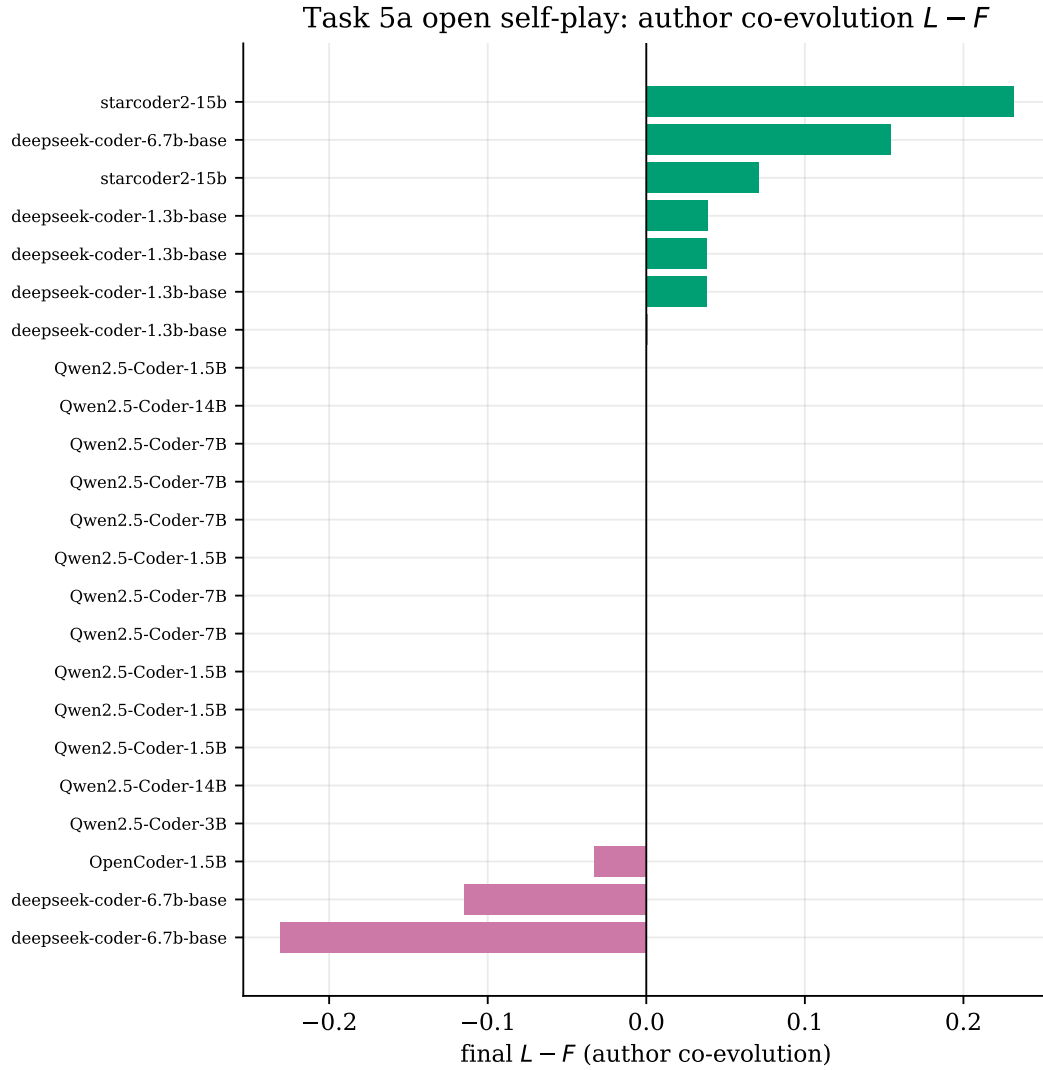


Figure 6: Final $L - F$ (live-author minus frozen-author, the self-referential signal) per open-weight model. > 0 means updating the author beyond a frozen-author escalating curriculum genuinely helps.

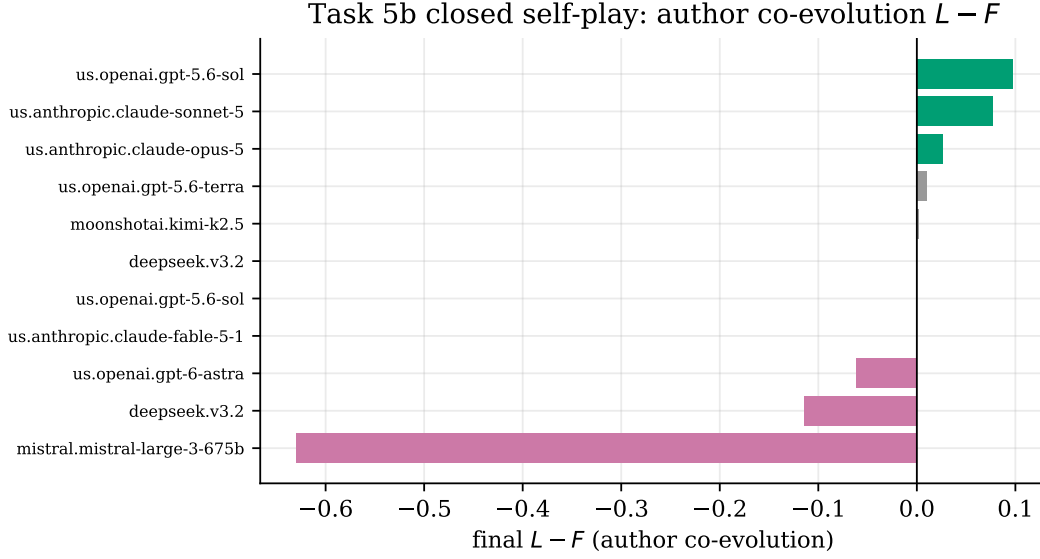


Figure 7: Final $L - F$ per closed/API model (procedure co-evolution channel). Most cluster near zero; early large spikes did not survive longer horizons.

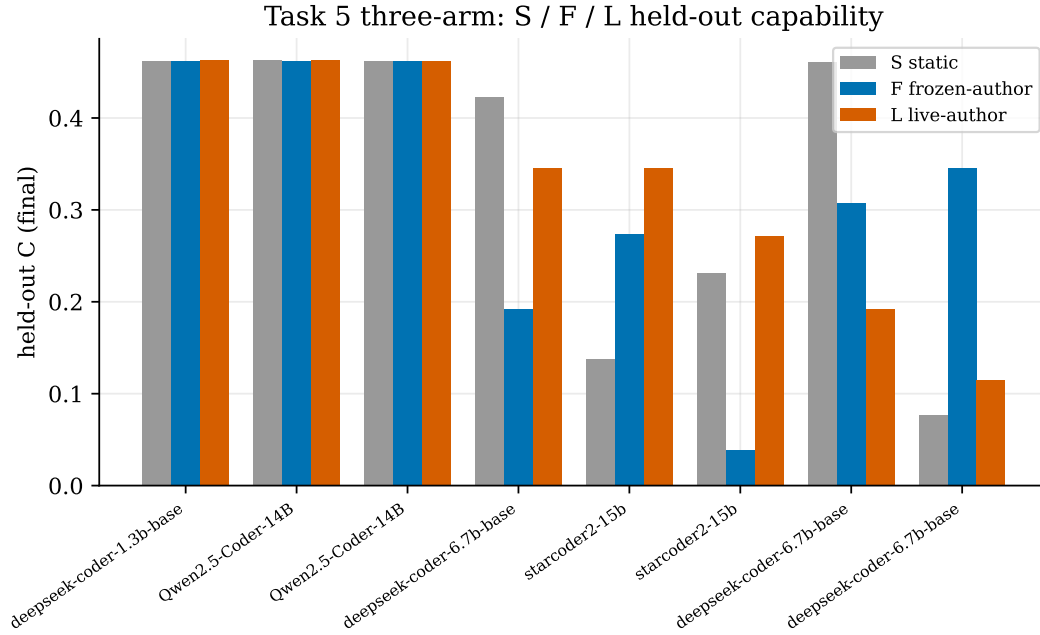


Figure 8: Three-arm self-play at equal budget on a fixed held-out ladder: STATIC (fixed frontier), FROZEN-AUTHOR (escalating frontier, frozen base author), LIVE-AUTHOR (co-evolving author). $F - S$ isolates the adaptive-curriculum benefit; $L - F$ isolates author co-evolution.

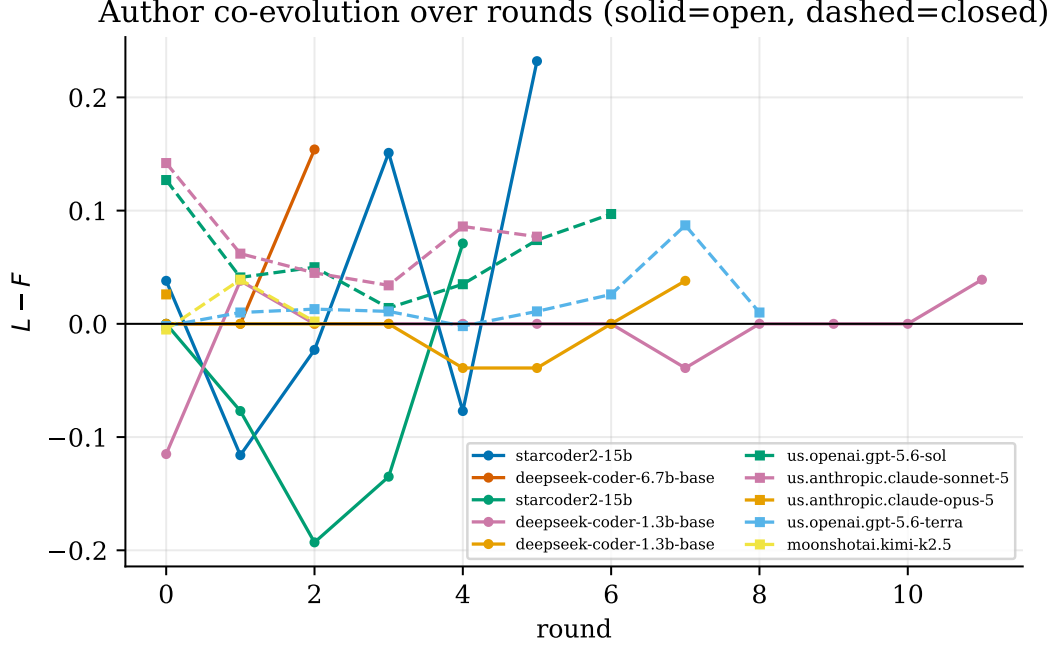


Figure 9: Per-round $L - F$ trajectories (solid = open-weight, dashed = closed/API). Long horizons reveal whether early positive spikes compound or regress to zero.

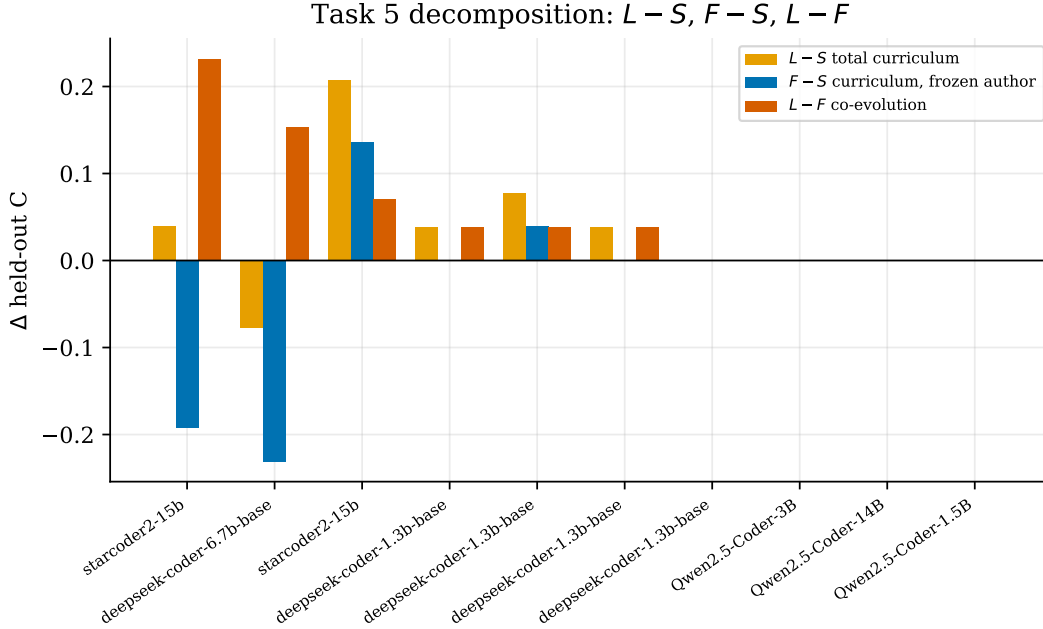


Figure 10: Decomposing the self-play benefit: $L - S$ (total adaptive-curriculum gain) = $(F - S)$ (curriculum with a frozen author) + $(L - F)$ (extra gain from co-evolving the author). Most models get their gain from $F - S$, not $L - F$.

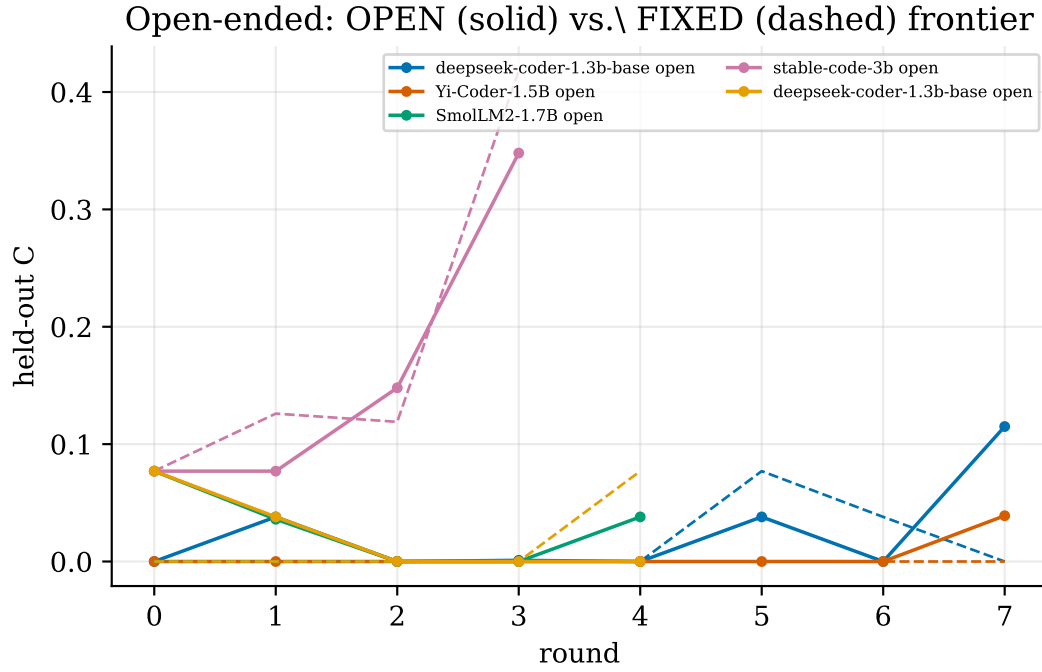


Figure 11: Open-ended RSI: a learner facing an OPEN escalating self-authored frontier (solid) vs. a FIXED static frontier (dashed) at equal budget. Overlap means open-endedness is a one-time upgrade, not compounding.

4 Task 4: closed→open

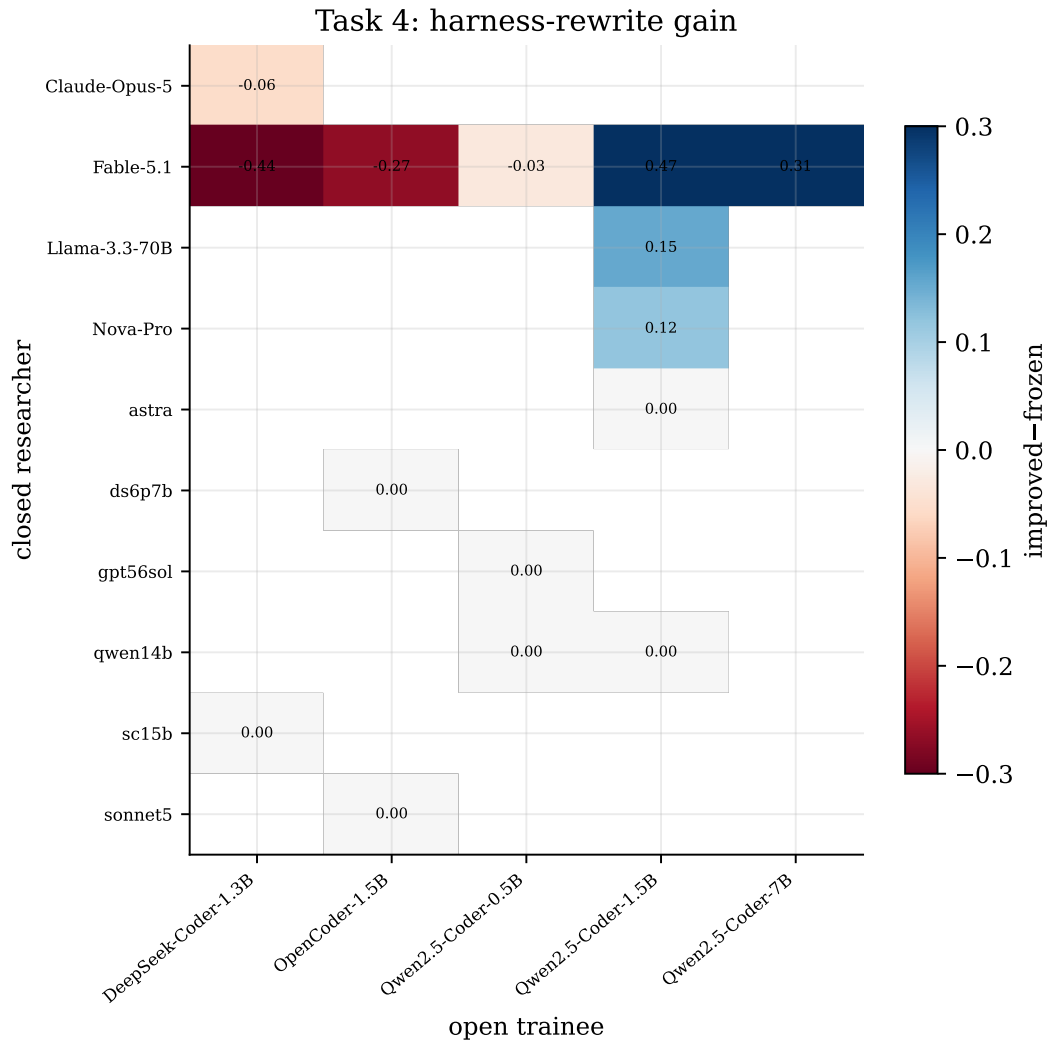


Figure 12: Task 4 (closed→open): a closed frontier researcher rewrites an open trainee’s training harness. Cell = improved - frozen held-out C on the trainee (positive = the rewritten harness helps).

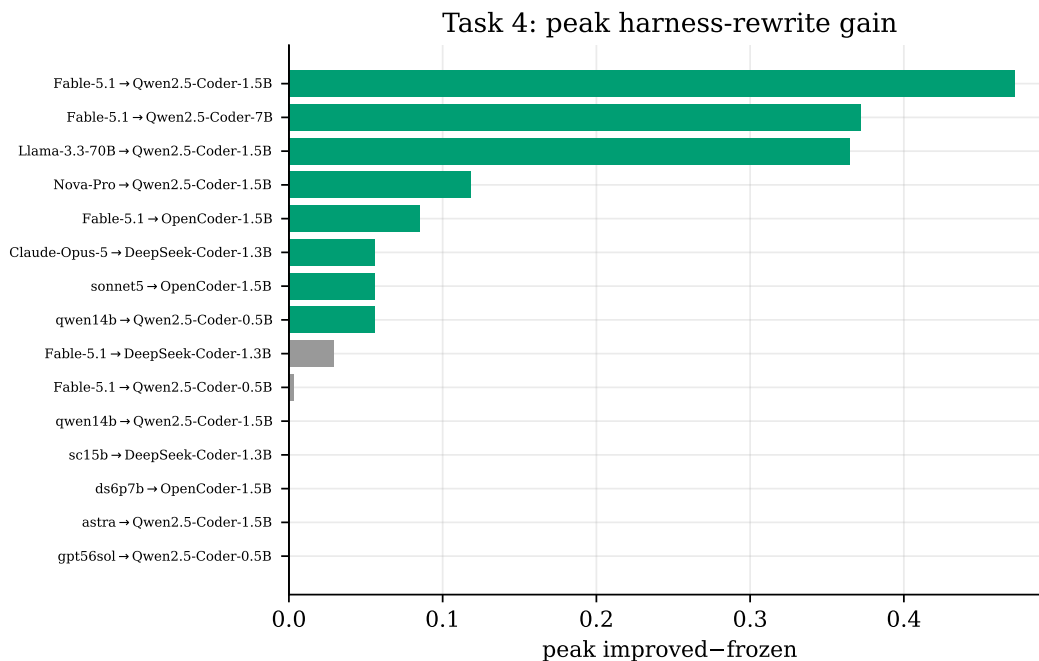


Figure 13: Peak improved-frozen over rounds for each researcher→trainee pair.

5 Task 3: procedure-RSI

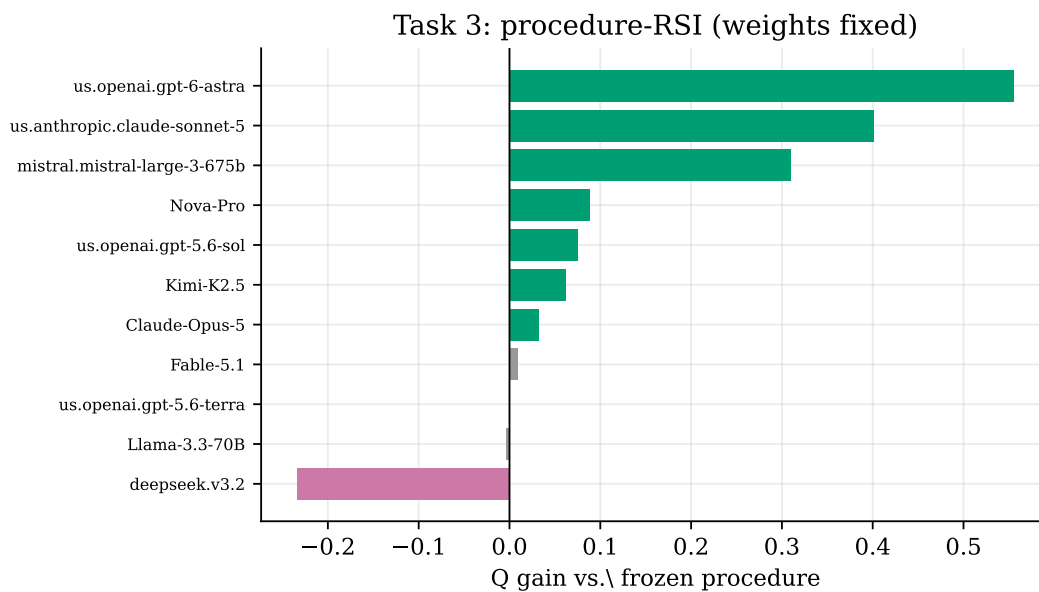


Figure 14: Task 3: a frozen-weight model rewrites its own strategy library + verified archive. Held-out Q gain vs. the frozen procedure; frontier models improve their own procedure substantially.

6 Comparators & weight-RSI

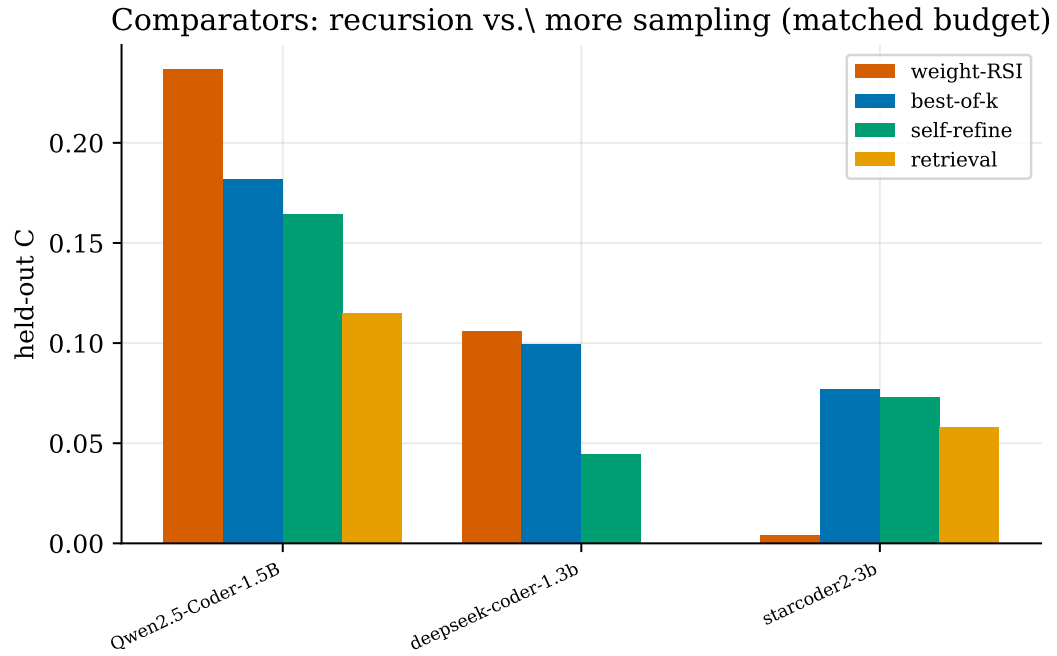


Figure 15: Is it really recursion? Weight-RSI vs. non-recursive controls (best-of-k, self-refine, retrieval) at matched generation budget. $\text{recursion_gain} = \text{weight-RSI} - \text{best comparator}$.

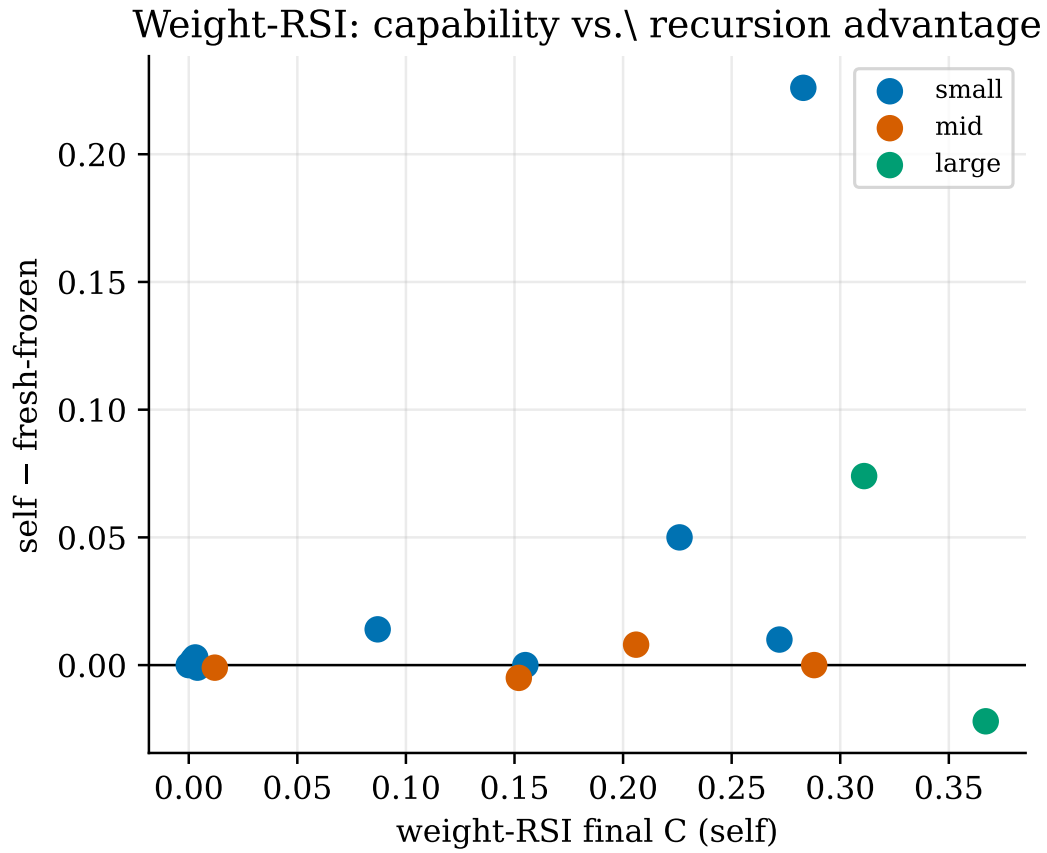


Figure 16: Weight-RSI: final self-trained held-out C vs. the self-minus-fresh-frozen advantage (positive = training on own kernels beats an equal-budget fresh baseline).

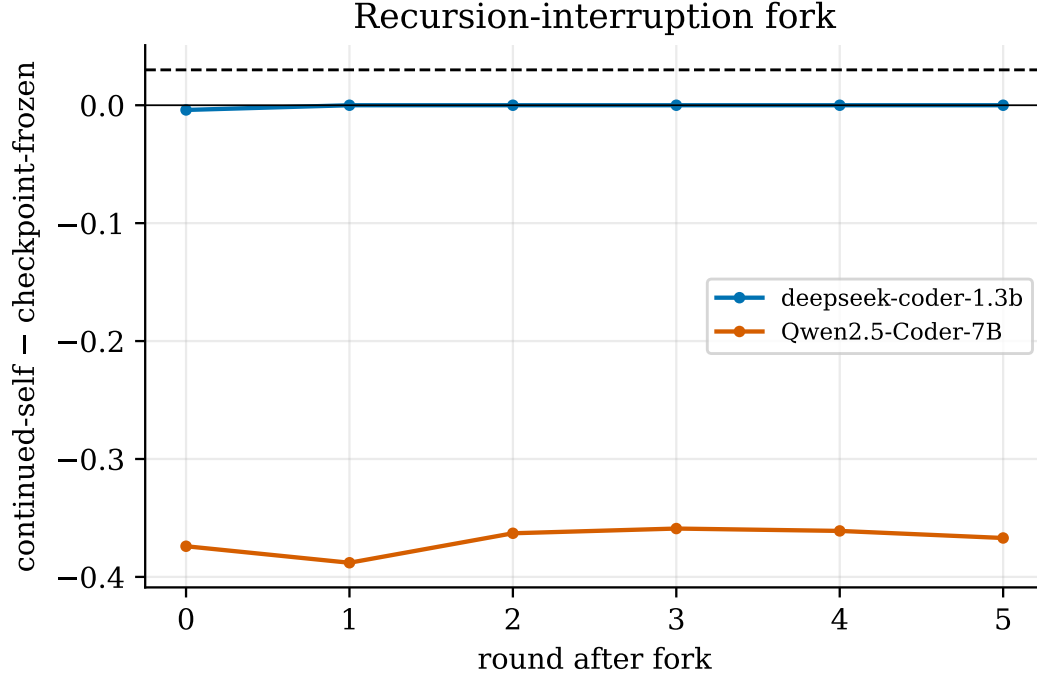


Figure 17: Recursion-interruption control: continued self-training vs. a checkpoint-frozen producer at equal budget. Sustained > 0 = genuine compounding rather than a one-time producer upgrade.

7 Flows & causality

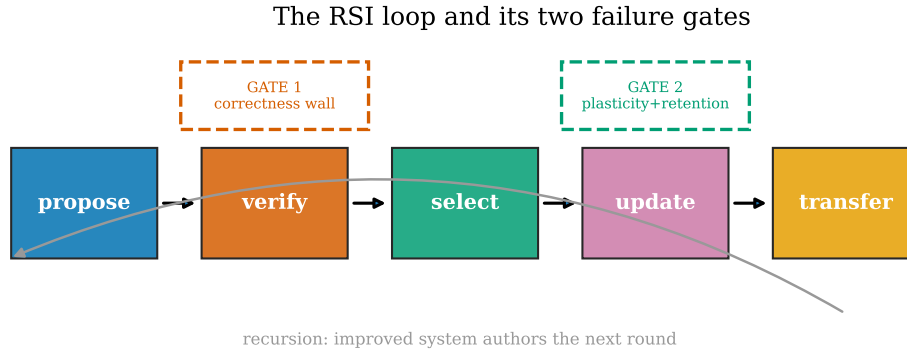


Figure 18: The recursive self-improvement loop (propose→verify→select→update→transfer) and the two gates our data identifies: Gate 1, the correctness wall (no verified kernel $\Rightarrow$ no gradient), and Gate 2, plasticity+retention (drift saturation / forgetting collapses compounding).

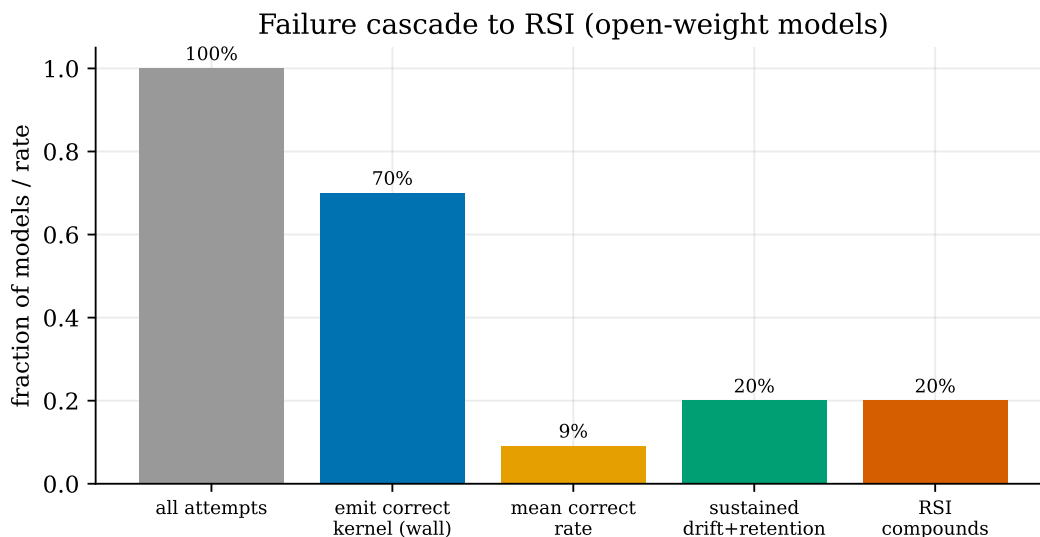


Figure 19: Attrition from attempt to compounding RSI across open-weight models: most attrition happens at the correctness wall (emitting any correct kernel) and again at sustained plasticity+retention.

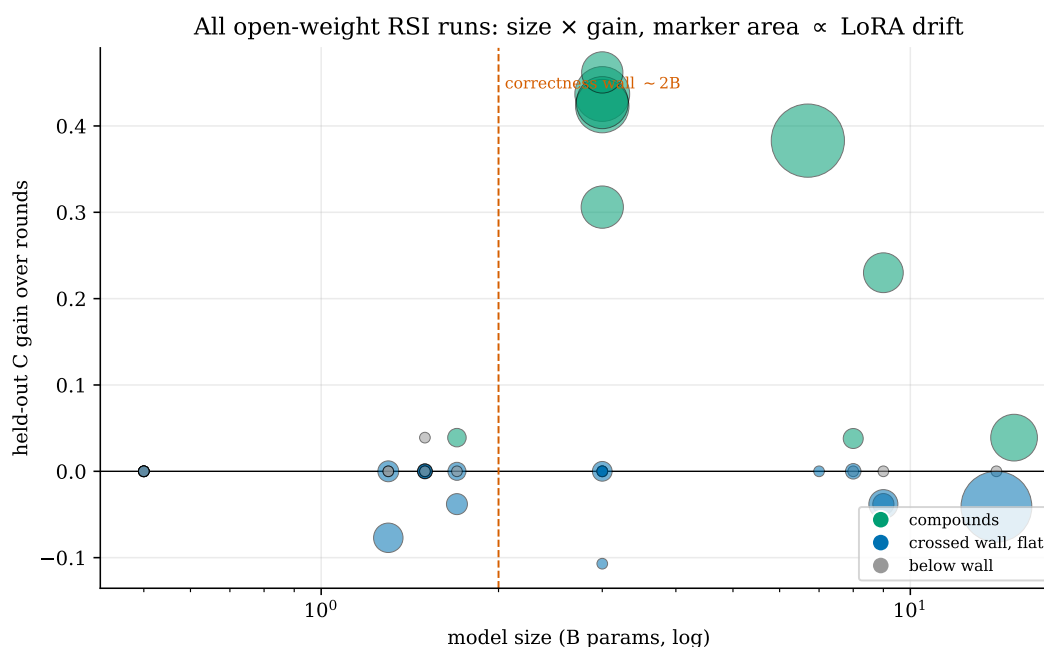


Figure 20: Every open-weight weight-RSI run positioned by model size (x) and held-out C gain (y); marker area $\propto$ sustained LoRA drift; color = outcome. The 2B correctness wall and the mid-scale compounding band are visible at a glance.

8 Large causality & internal-failure graphs

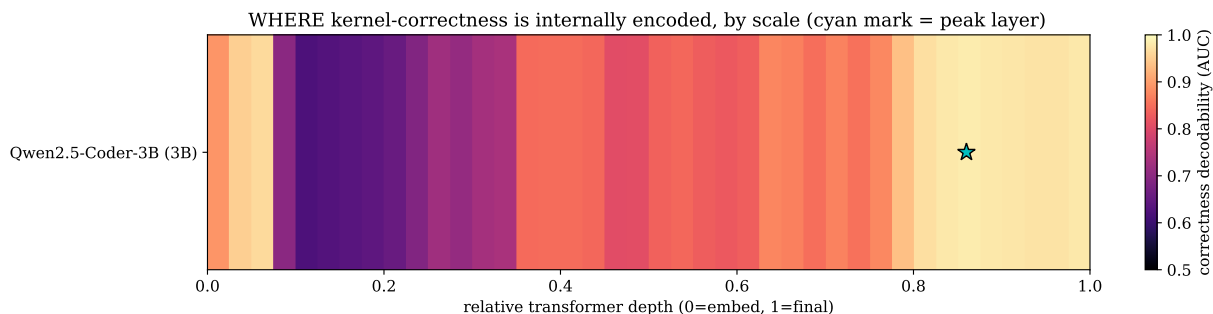


Figure 21: Per-layer linear decodability of kernel correctness (AUC of a correct-vs-incorrect probe) across relative transformer depth, one row per model sorted by scale; ★ marks the peak layer. Correctness becomes cleanly decodable in mid-to-late layers – the model internally represents which kernels are correct. Rows populate as GPU interp probes land.

Internal-failure causality DAG: each model's path Scale → Wall → Gradient → Drift → Retention → Diversity → Outcome (50 models; edge width & marker area $\propto$ LoRA drift)

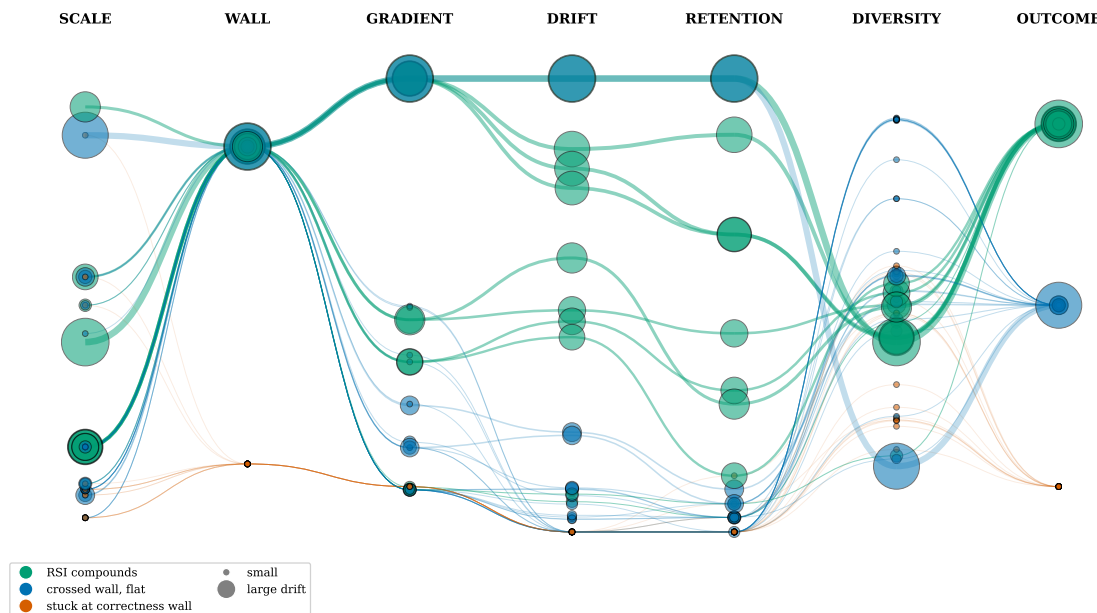


Figure 22: Every open-weight run traced through the causal chain from scale to RSI outcome, in lanes with translucent curved edges (marker area and edge width $\propto$ sustained LoRA drift, color = outcome). Compounders (green) form a bright high-drift/high-retention bundle; models stuck at the correctness wall (orange) die at the second lane with near-zero drift downstream.

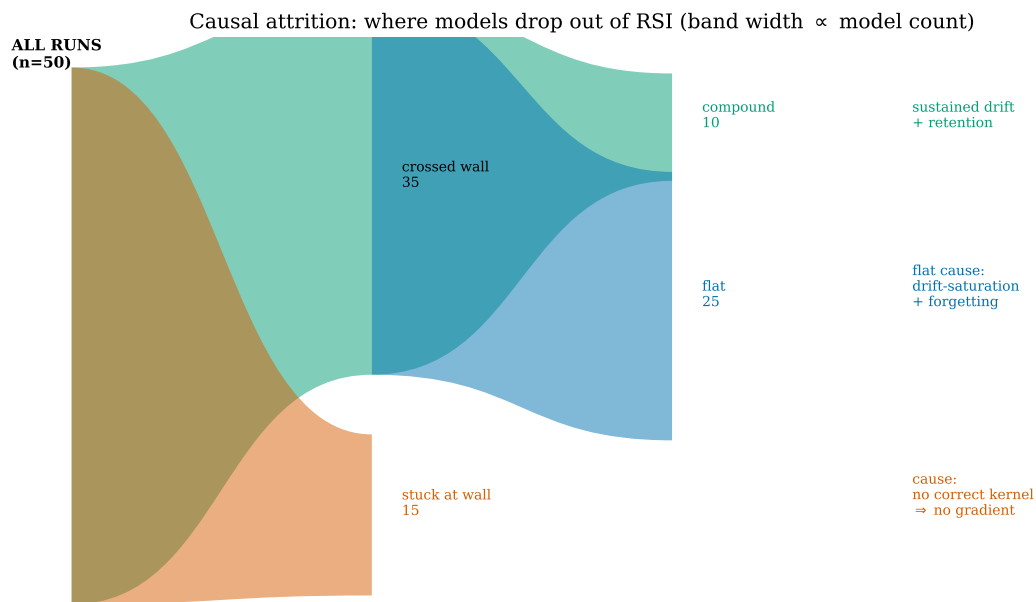


Figure 23: Causal attrition of 50 open-weight runs: most are lost at Gate 1 (the correctness wall – no verified kernel, no gradient); among wall-crossers only a minority compound, the rest going flat via drift-saturation and forgetting (Gate 2).

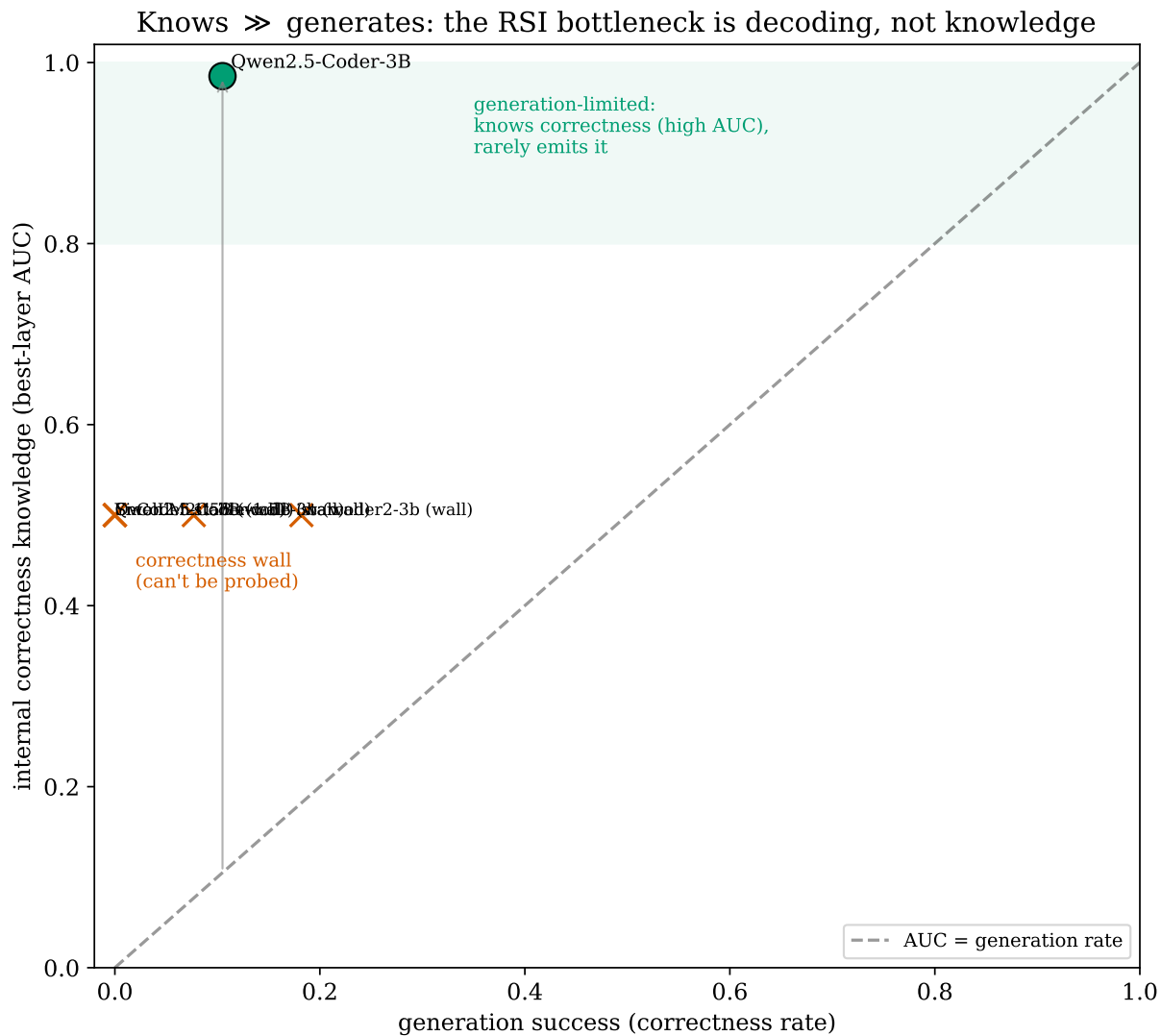


Figure 24: Internal correctness knowledge (best-layer probe AUC) vs. actual generation success. Points far above the diagonal are *generation-limited* – the model represents correctness internally (AUC $\approx$ 0.98) yet emits a correct kernel only a small fraction of the time; models at the wall cannot be probed at all.

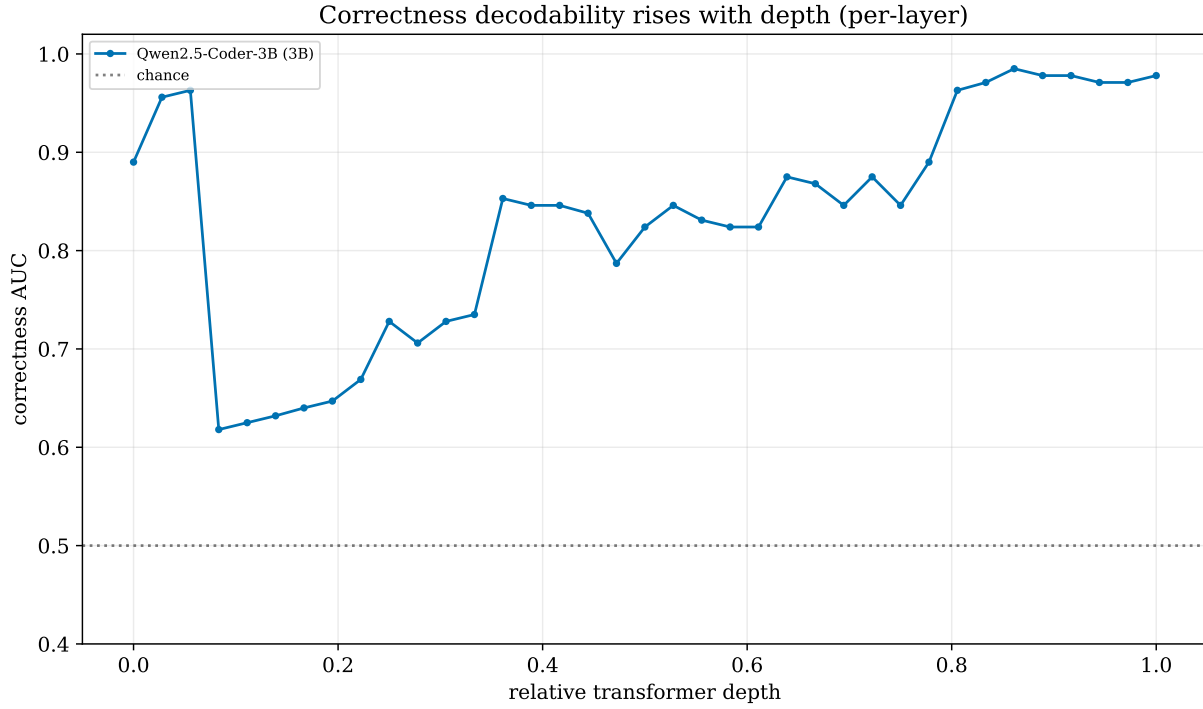


Figure 25: Layer-wise correctness decodability vs. depth: the correct-vs-incorrect signal is weak at the embedding and sharpens through mid/late layers, indicating correctness is a computed, decodable feature rather than a surface one.

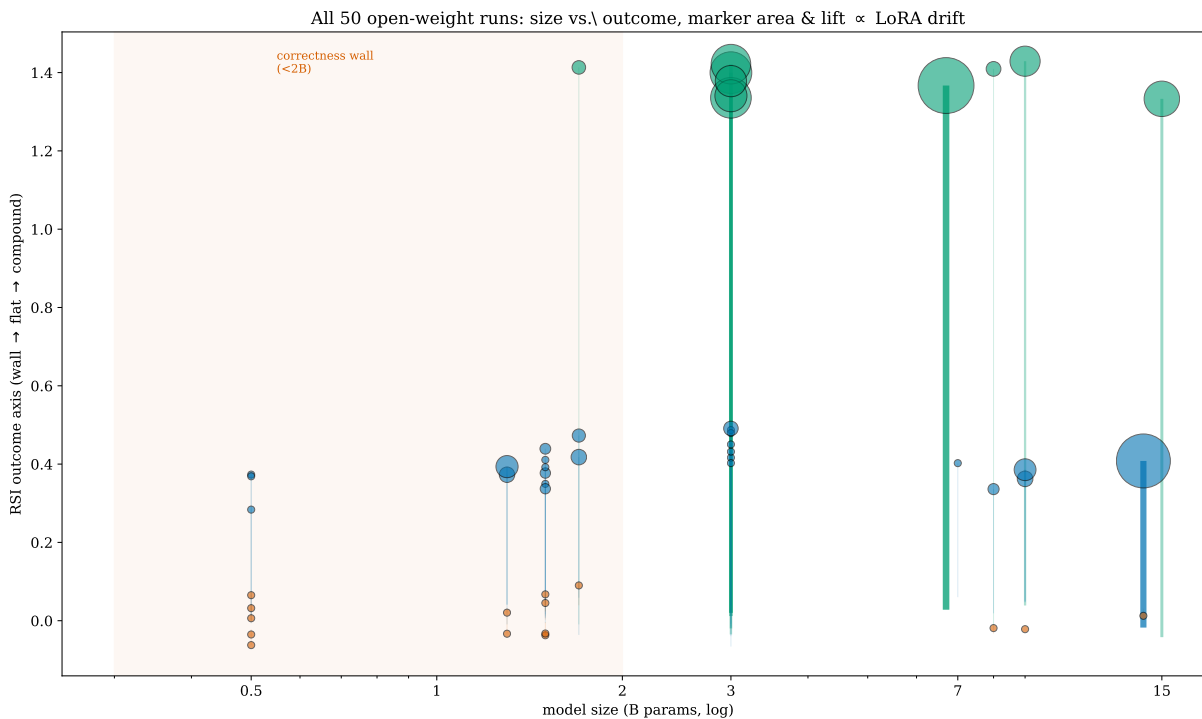


Figure 26: Dense view of every open-weight run: horizontal position = scale, vertical = RSI outcome, marker area and the base→self connector width $\propto$ sustained LoRA drift. The sub-2B correctness wall (shaded) and the mid-scale high-drift compounding band emerge from the raw point cloud.

RSI attrition funnel across open-weight models

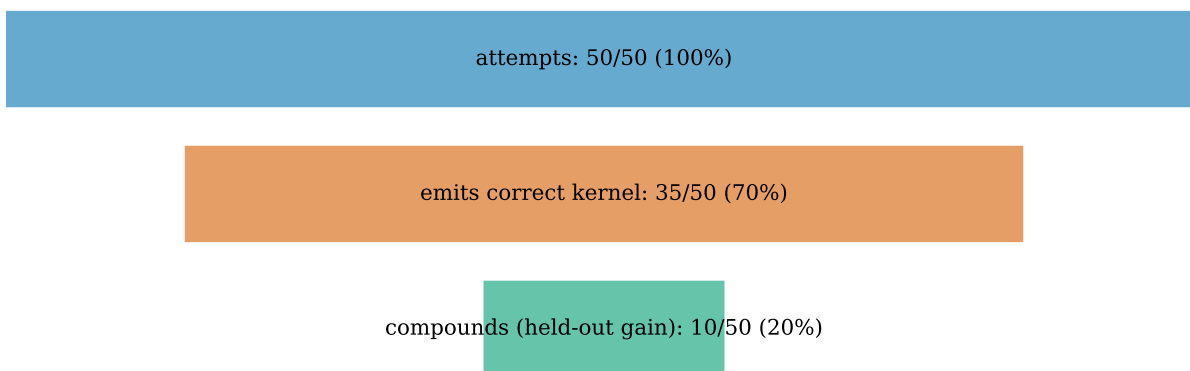


Figure 27: Attrition funnel across open-weight models: the dominant drop is at emitting any verified-correct kernel (the correctness wall); a further drop separates one-shot correctness from sustained held-out compounding.

9 Gallery — dense graphs

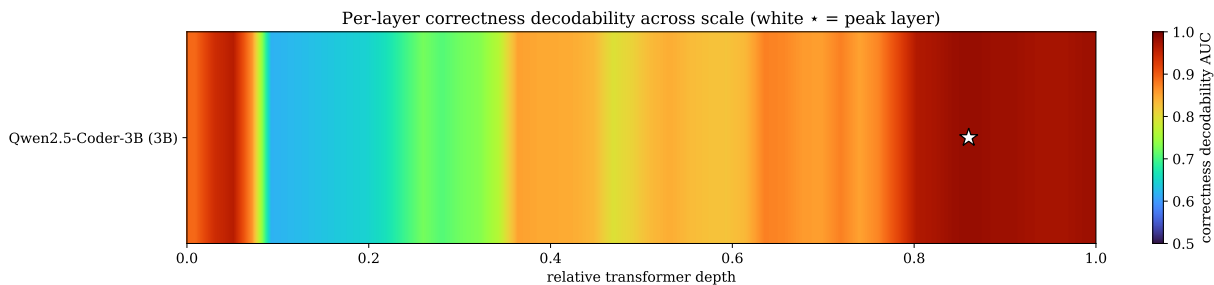


Figure 28: Full-resolution model×layer heatmap of how linearly decodable kernel-correctness is at each transformer depth (interpolated to a common grid, sorted by scale). The decodable-correctness signal concentrates in mid/late layers; it fills in as more interp probes land.

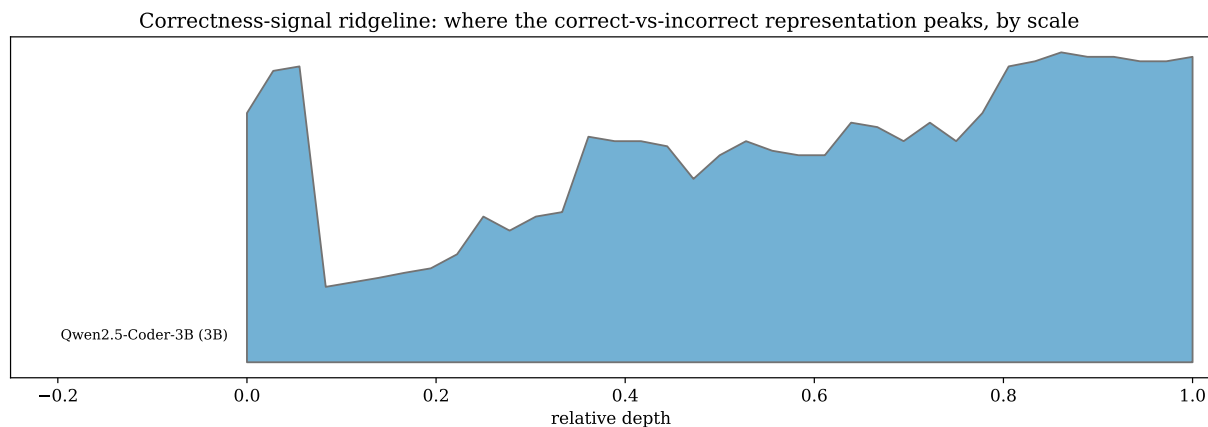


Figure 29: Ridgeline of per-layer correctness AUC (each curve a model, stacked by scale). Shows the internal correctness “wave” and at what depth it crests as models grow.

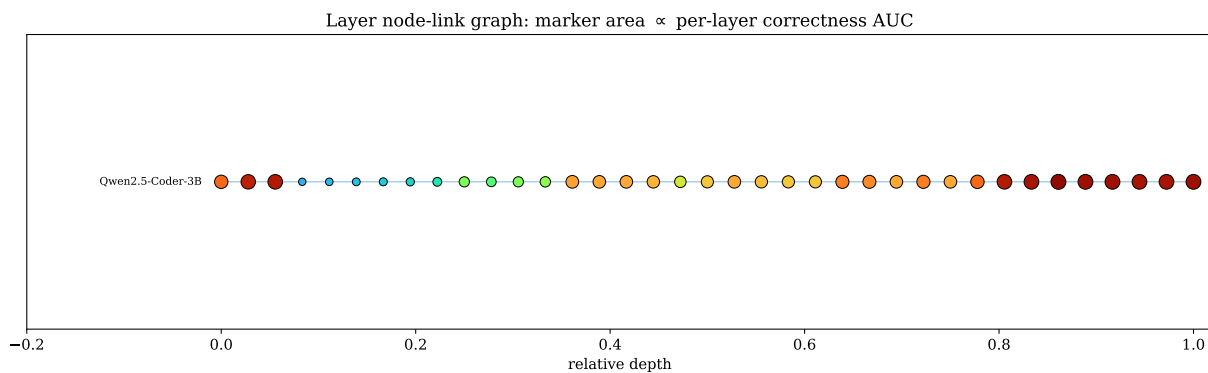


Figure 30: Node-link view of the correctness signal: one node per (model, layer), area and colour $\propto$ decodability AUC, translucent edges tracing depth. Dense bright bands mark the depth range where correctness becomes linearly readable.

Mechanism-metric correlation chords (green +, orange -; width $\propto |r|$) across 50 runs

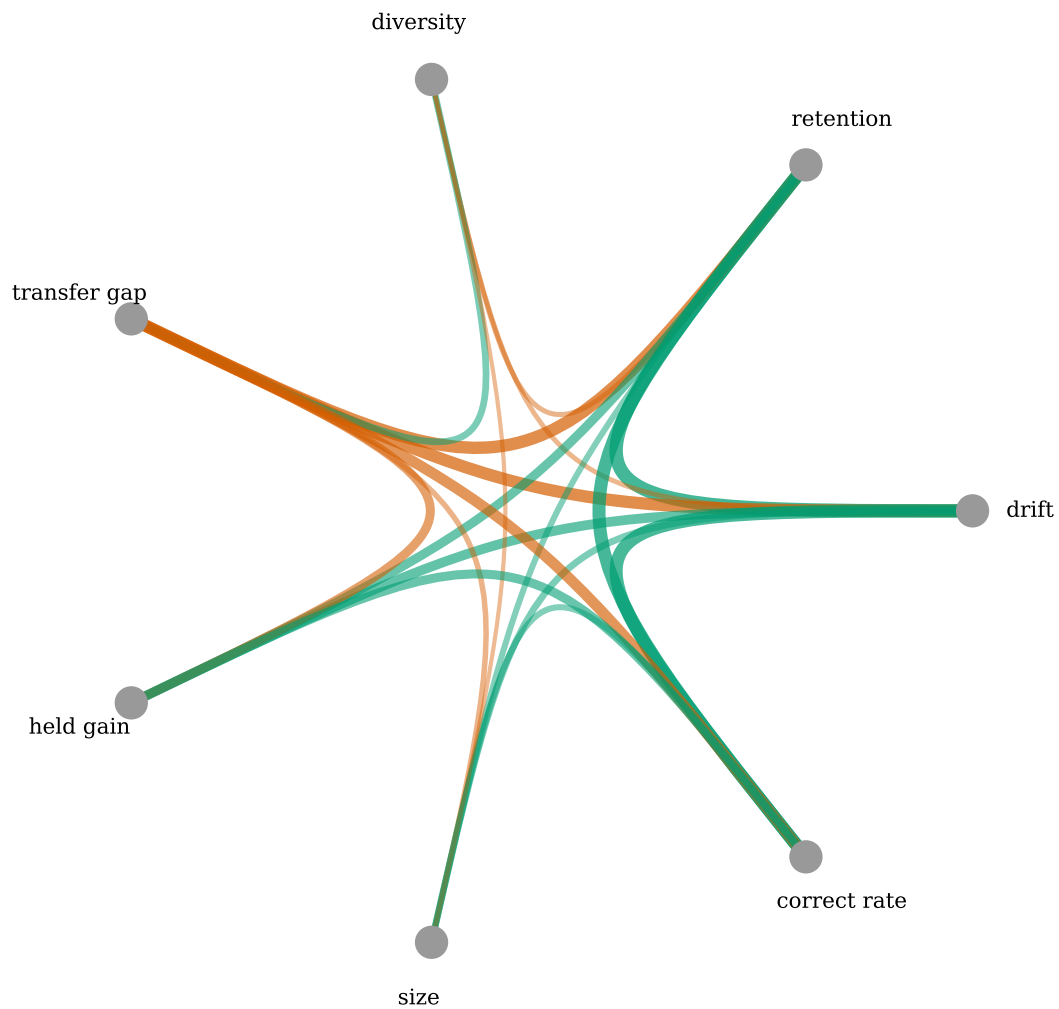


Figure 31: Circular correlation graph of mechanism metrics across all open-weight runs. Sustained drift and retention co-vary with held-out gain (green chords); transfer-gap and diversity relate oppositely – the internal fingerprint of compounding.

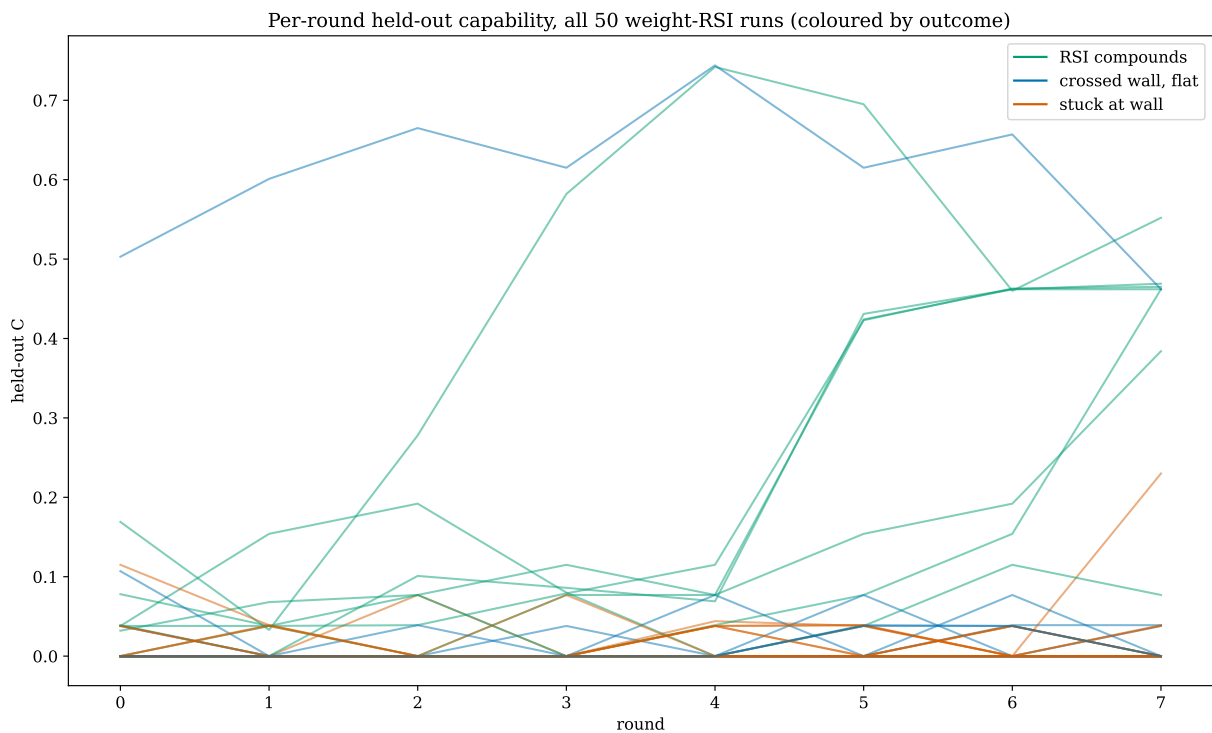


Figure 32: Every weight-RSI run’s held-out capability over rounds, coloured by outcome. Compounders separate upward; wall-stuck runs stay pinned at zero – the dynamics behind the static leaderboard.

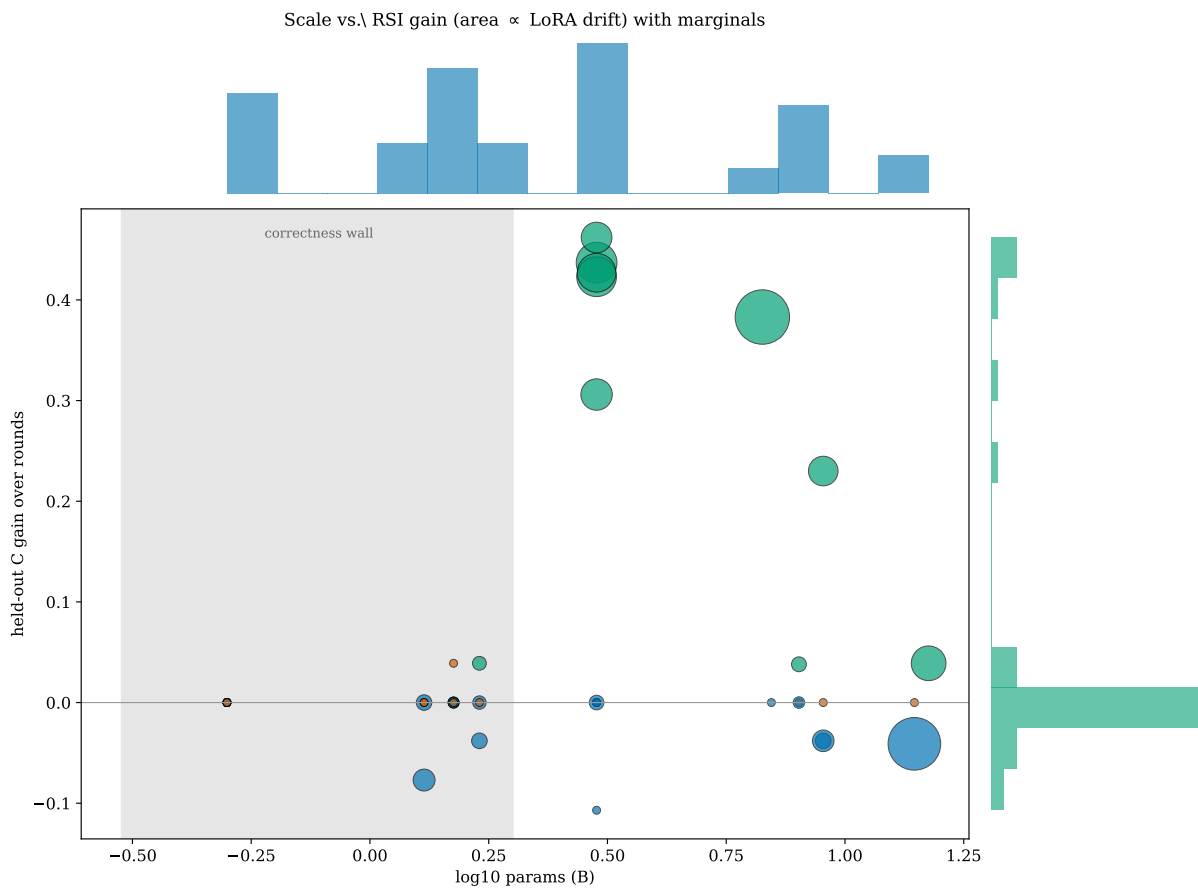


Figure 33: Every run positioned by scale (x) and held-out gain (y), bubble area $\propto$ sustained drift, colour = outcome, with marginal histograms. The sub-2B wall and the mid-scale compounding band are visible in the joint and marginal densities.

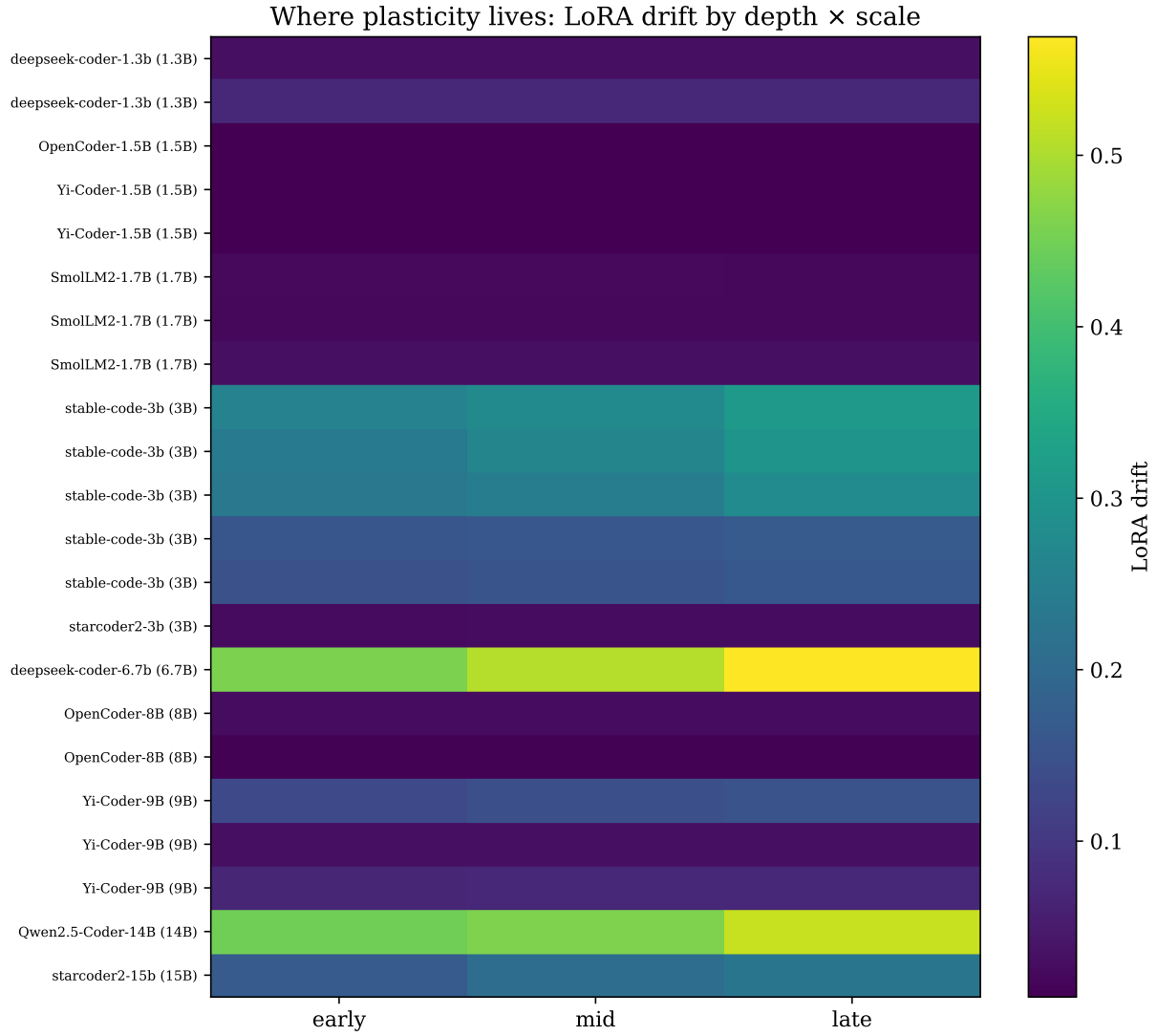


Figure 34: LoRA weight drift by transformer block (early/mid/late) for every run, sorted by scale – where in the network self-training actually moves weights, and how that shifts with size.

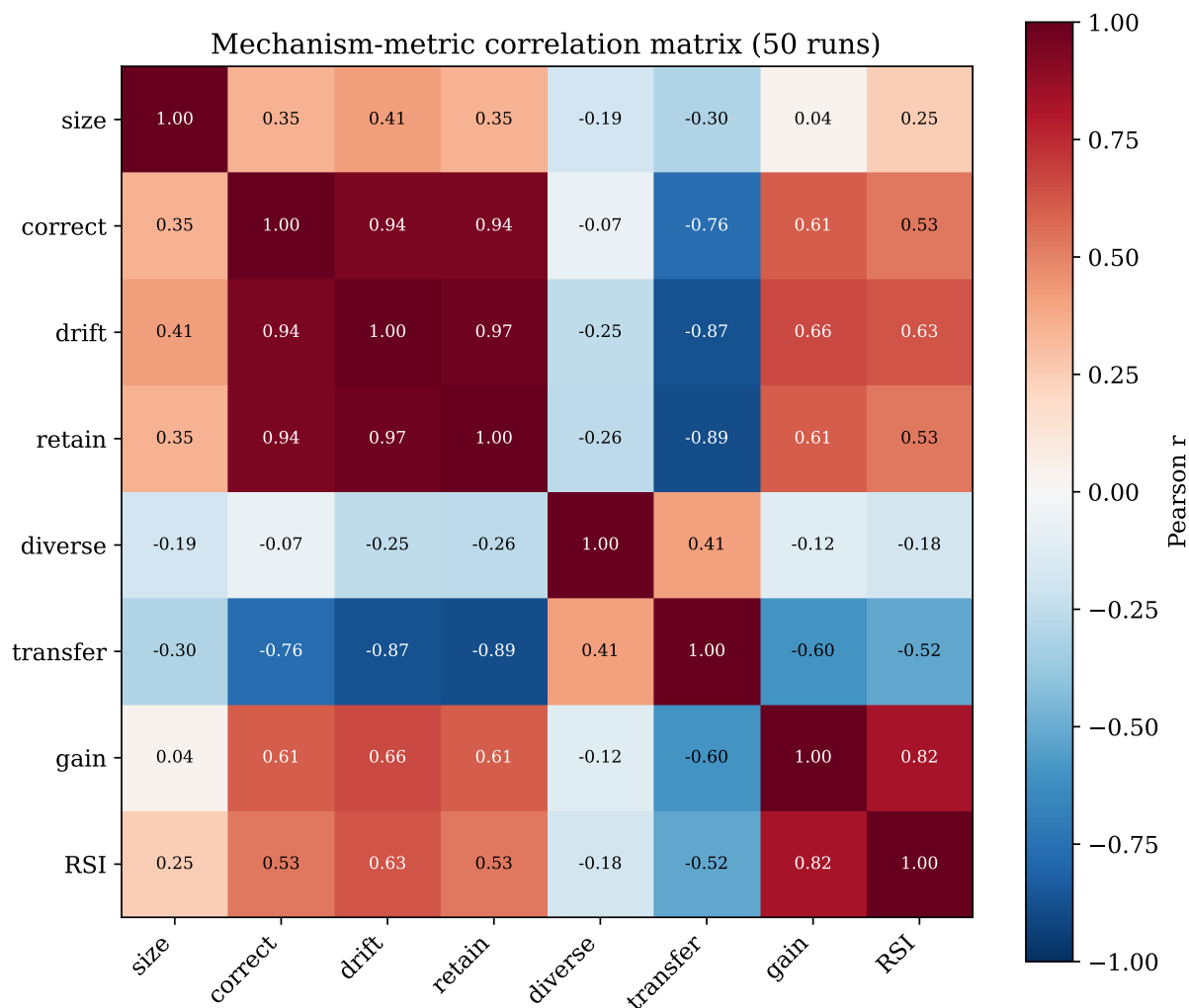


Figure 35: Correlation matrix of mechanism metrics and RSI outcome across all runs – drift and retention are the strongest positive correlates of held-out gain; the correctness-rate gate dominates.

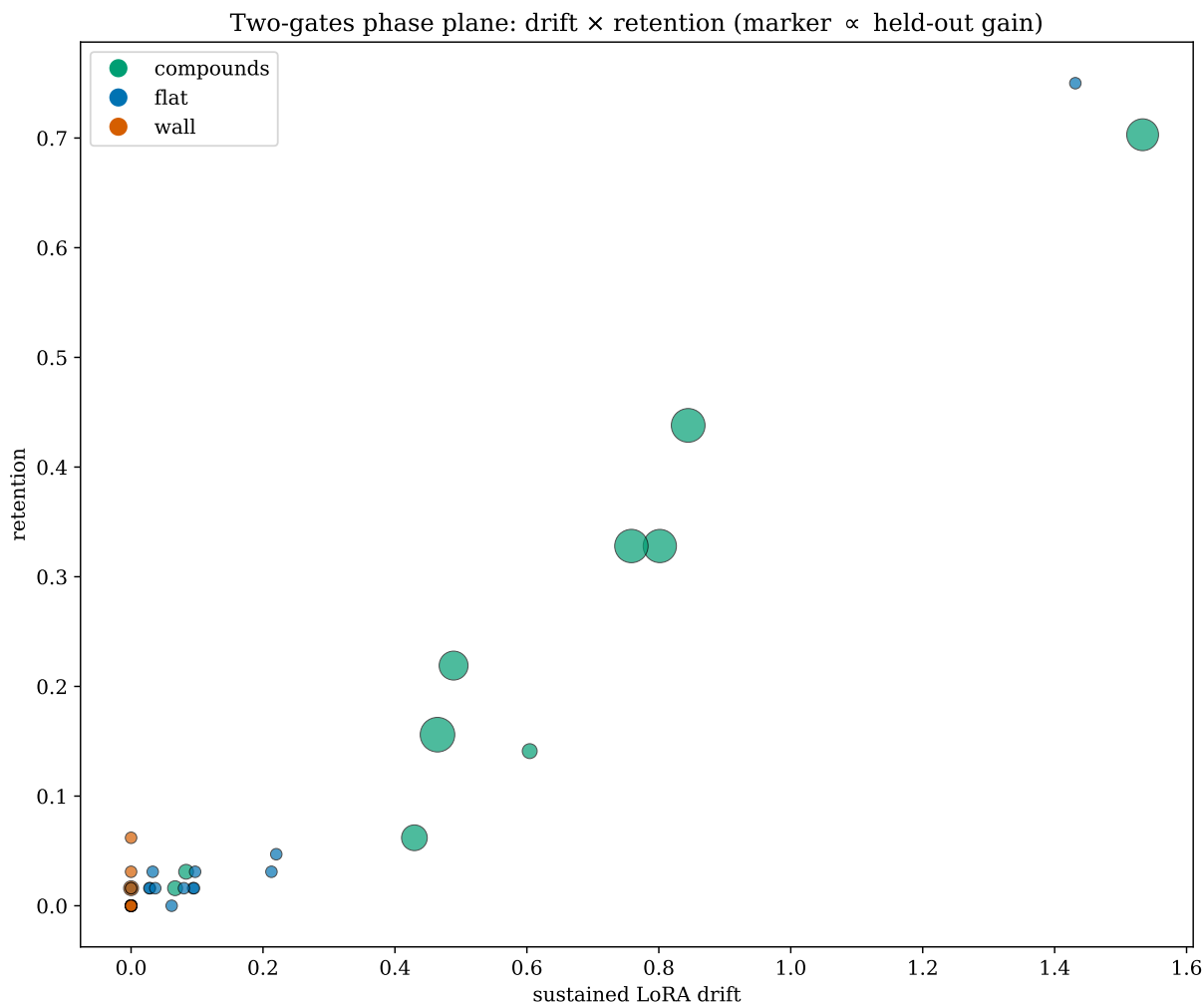


Figure 36: Phase plane of the second gate: compounders occupy the high-drift/high-retention corner; flat runs cluster at low drift (saturation) or low retention (forgetting). Marker area $\propto$ held-out gain.

Task 5 author co-evolution $L - F$ per round, every self-play run



Figure 37: Per-round $L - F$ (live- minus frozen-author) for every open and closed self-play run. Most hover at zero (adaptive-curriculum only); a few sustain > 0 (genuine author co-evolution) – the headline metric, run by run.

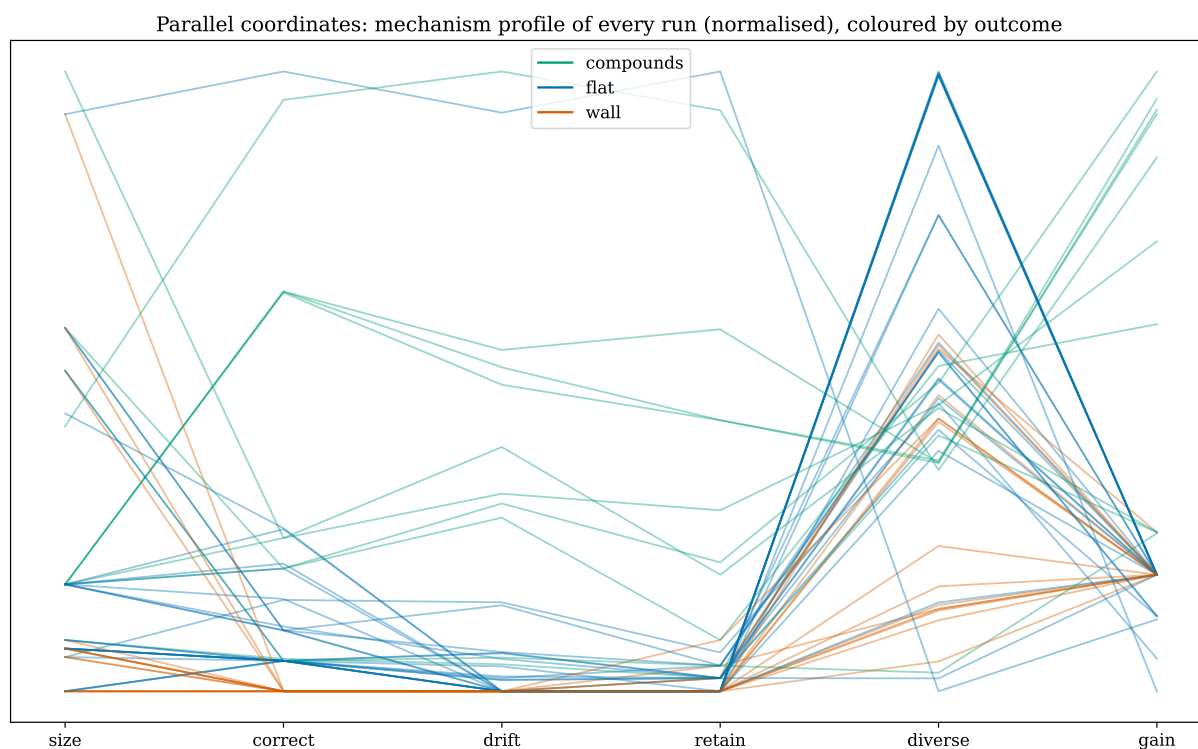


Figure 38: Parallel-coordinates fingerprint of every run across the mechanism axes, coloured by outcome – compounders share a high-correct/high-drift/high-retention profile that the flat and wall runs lack.

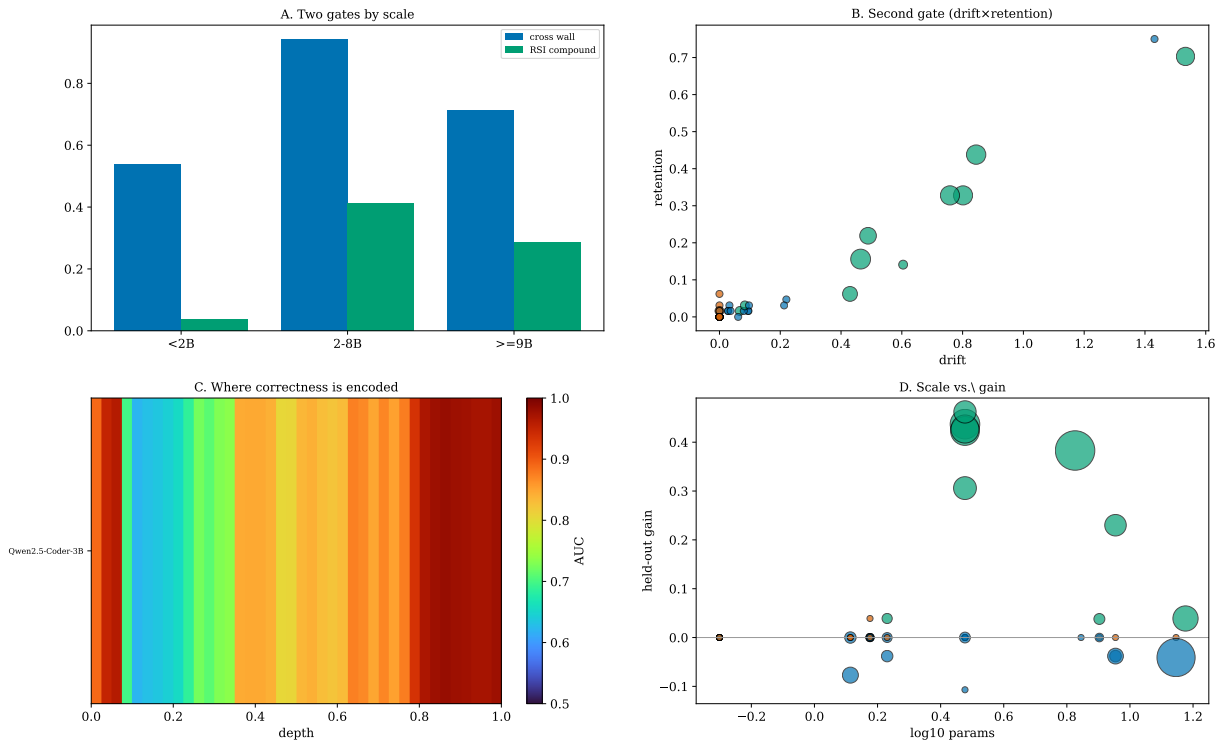


Figure 39: Single-figure synthesis: (A) the two scale gates, (B) the drift×retention second gate, (C) where correctness is internally encoded by depth, (D) scale vs. held-out gain. The full mechanistic story of the benchmark in one poster.

10 Probe-as-intervention (self-verification)

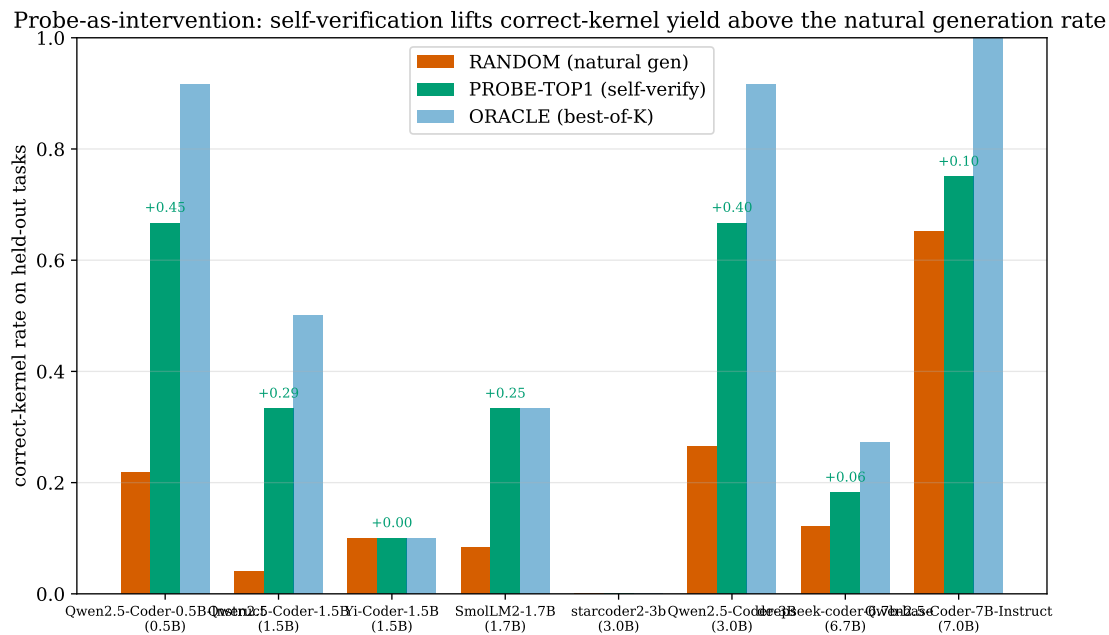


Figure 40: Self-verification (probe-guided selection) vs. the model’s natural single-draw success (RANDOM) and the ORACLE best-of- K upper bound, at equal generation budget K , on held-out tasks. Annotated $\text{+lift} = \text{PROBE-TOP1} - \text{RANDOM}$. Probe selection recovering much of the oracle gap turns the correlational correctness-AUC into a causal decode-time intervention.

K-sweep: probe-guided selection vs.\ random — does self-verification lift models over the correctness wall?

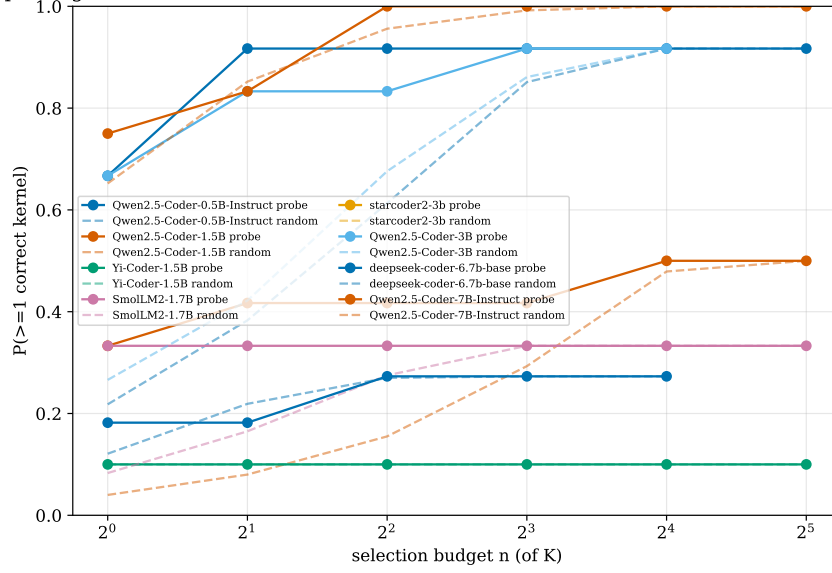


Figure 41: For each model, probability of obtaining ≥ 1 correct kernel when keeping the top- n of K candidates by probe score (solid) vs. a random n (dashed). A probe curve rising above random — especially lifting a sub-2B model from ≈ 0 — is direct evidence that the correctness wall is partly a decode/selection artifact.